The proof requires calculation of the vector fields

$$\boldsymbol{u}_0 = \boldsymbol{\Pi f}, \qquad \boldsymbol{u}_1 = \boldsymbol{\Pi}\,[\boldsymbol{f},\boldsymbol{g}]\,, \qquad \boldsymbol{u}_2 = \boldsymbol{\Pi}\,[\boldsymbol{f},[\boldsymbol{f},\boldsymbol{g}]], \qquad \boldsymbol{u}_3 = \boldsymbol{\Pi}\,[\boldsymbol{f},[\boldsymbol{f},[\boldsymbol{f},\boldsymbol{g}]]],$$

where $\boldsymbol{\Pi}$ is the projector onto the first three components. Because $\boldsymbol{g} = \sigma \boldsymbol{e}_4$, it is enough to show that any pair from among these four vectors are linearly independent, except possibly on $\mathcal{M}_0$. Note that projection can only be performed after the full calculation of the relevant commutator. Because the commutators quickly become quite messy, we used a symbolic computing package to assist in these calculations.

The first two vectors are:

$$\mathbf{u}_0 = p\partial_q - (q + \lambda p)\partial_p - (\lambda\xi_1 + \xi_1\xi_2)\partial_{\xi_1}, \tag{8.48}$$

and

$$\mathbf{u}_1 = \sigma p \frac{\partial\lambda}{\partial\xi_2}\partial_p + \xi_1\left(1 + \frac{\partial\lambda}{\partial\xi_2}\right)\partial_{\xi_1}. \tag{8.49}$$

After expanding the derivatives of λ we have

$$\mathbf{u}_1 = p\partial_q + \xi_1^2\left(\frac{-q + p\xi_2}{D}\right)\partial_p - 2\xi_1 p\left(\frac{-q + p\xi_2}{D}\right)\partial_{\xi_1}, \tag{8.50}$$

and

$$\mathbf{u}_2 = -\sigma(\xi_1^2 p/D)\partial_p + (2\sigma\xi_1 p^2/D)\partial_{\xi_1}, \tag{8.51}$$

where $D = 2p^2 + \xi_1^2$.

If $q \neq 0$ and $p \neq 0$, then clearly $\mathbf{u}_1$ and $\mathbf{u}_2$ are linearly independent. However, if $p = 0$ then $\mathbf{u}_2 = 0$. We proceed to compute the commutator $[\boldsymbol{f},[\boldsymbol{f},\boldsymbol{g}]]$. Projecting to the first three components results in

$$\mathbf{u}_2 = \sigma\left(\frac{\xi_1^2 p}{D}\right)\partial_q - \sigma\left(\frac{2\gamma p^3 + \gamma\xi_1^2 p - \xi_1^2 q}{D^2}\right)\partial_p + 2\sigma\xi_1 p\left(\frac{2\gamma p^3 + \gamma\xi_1^2 p - \xi_1^2 q}{D^2}\right)\partial_{\xi_1}. \tag{8.52}$$

Again, we substitute $p = 0$ and find that

$$\mathbf{u}_2|_{p=0} = (\sigma\xi_1^2 q/D^2)\partial_p. \tag{8.53}$$

Unfortunately, this is parallel to $\mathbf{u}_1$ (for $p = 0$), and hence yet one more commutator is required in this case.